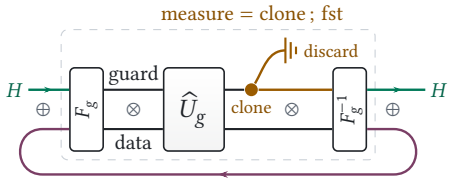
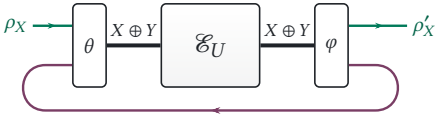
(a) the term: the guard is measured in every round



(b) the same loop as a trace of channels

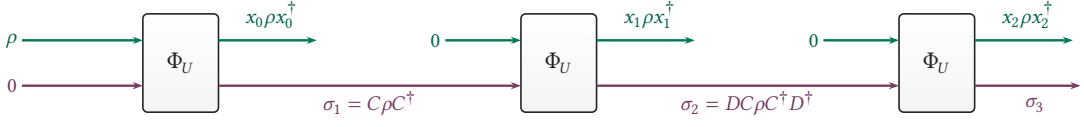


outside θ and φ: a pair of states, no cross terms;
inside: one state of the Hilbert sum $X \oplus Y$;

$$\theta(\rho_X, \rho_Y) = \begin{pmatrix} \rho_X & 0 \\ 0 & \rho_Y \end{pmatrix}, \quad \mathscr{E}_U \rho = U \rho U^\dagger,$$

φ keeps the two diagonal blocks

(c) Unroll $\Phi_U = \varphi \mathscr{E}_U \theta$ on pairs of positive operators



0 is the zero operator. The green outputs add in $\mathrm{End}(X)$; they carry no off-diagonal count terms.